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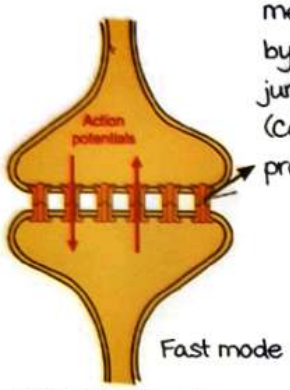
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SYNAPTIC TRANSMISSION AND NEUROTRANSMITTERS

Synaptic Transmission

00:00:48

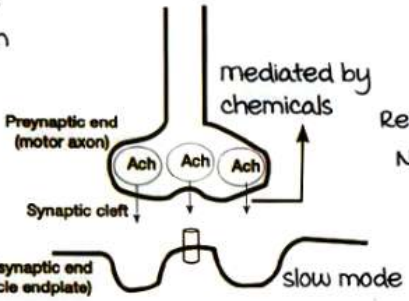
TYPES :



mediated by gap junctions (Connexin protein)

Fast mode

1. Electrical mode

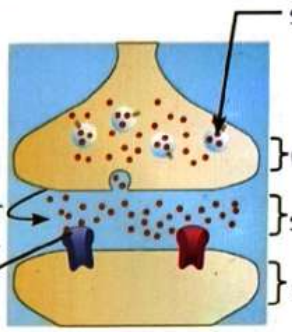


mediated by chemicals

slow mode

a. Neurotransmitters (NT)

STRUCTURE :



Stored NT

Presynaptic neuron

Release of NT

NT Receptor binding

Synaptic cleft

Postsynaptic neuron

Postsynaptic potential/end plate potential (Local potential)

↓

Undergoes summation

↓

Crosses threshold

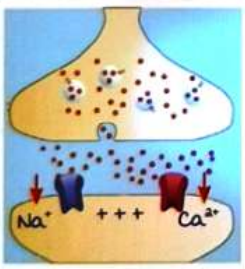
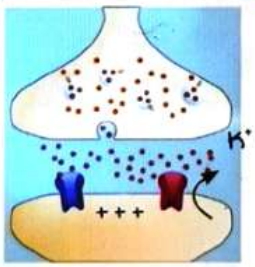
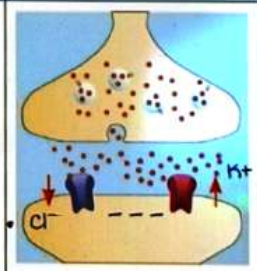
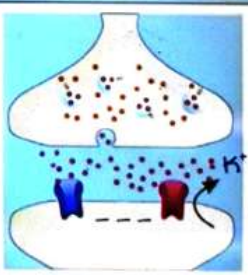
↓

Action potential generated

Note : Types of Synapse

- Axon – Axon.
- Axon – somatic.
- Axon – dendritic.

Post Synaptic Potentials (PSP) :

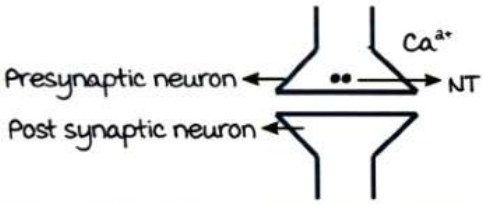
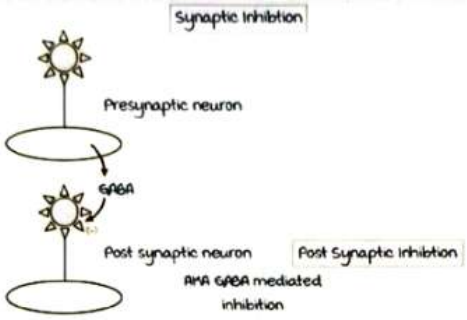
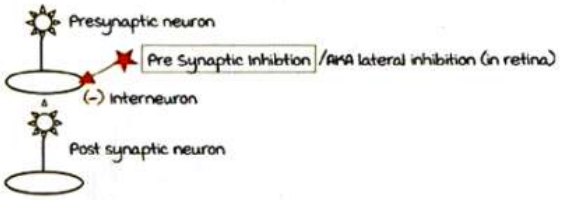
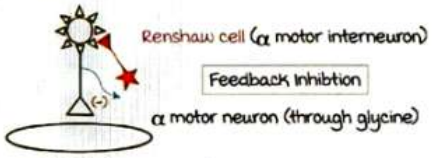
	Excitatory post synaptic potential		Inhibitory post synaptic potential	
	Fast EPSP	Slow EPSP	Fast IPSP	Slow IPSP
		 (Decrease efflux)		 (Increase efflux)
Example	Glutamate, acetylcholine	Autonomic ganglia, cardiac muscle, smooth muscle	GABA	
Latency	–	500 ms	–	500 ms



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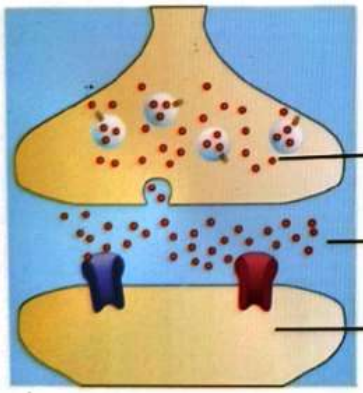
Regulation :

----- Active space -----

Facilitation	Inhibition
Ca^{a+} channel on presynaptic membrane ↓ Channel open for long time ↓ ↑↑↑ Ca^{a+} release ↓ ↑ neurotransmitter release ↓ Synaptic facilitation 	1. Synaptic inhibition  Presynaptic neuron GABA Post synaptic neuron Post Synaptic inhibition AKA GABA mediated inhibition 2. Presynaptic neuron  Pre Synaptic inhibition /AKA lateral inhibition (in retina) (-) interneuron Post synaptic neuron 3. Renshaw cell (α motor interneuron)  Feedback inhibtion α motor neuron (through glycine)

Neurotransmitters (NT) 00:24:00

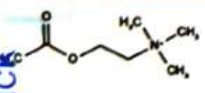
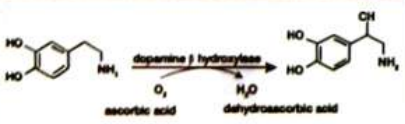
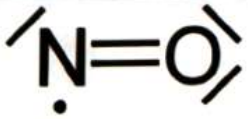
1st neurotransmitter discovered : Acetylcholine.



Criterias for NT :

1. NT synthesized & stored in presynaptic neuron
2. NT released into synapse
3. Post-synaptic neuron receptor

Classification :

CLASS I	CLASS II	CLASS III	CLASS IV
 : Acetylcholine	 Biogenic amines (Eg : Dopamine, norepinephrine, serotonin)	 Amino acids (Eg : Glutamate, GABA, glycine)	 Gases (Eg : Nitric oxide, carbon monoxide)

Feedback

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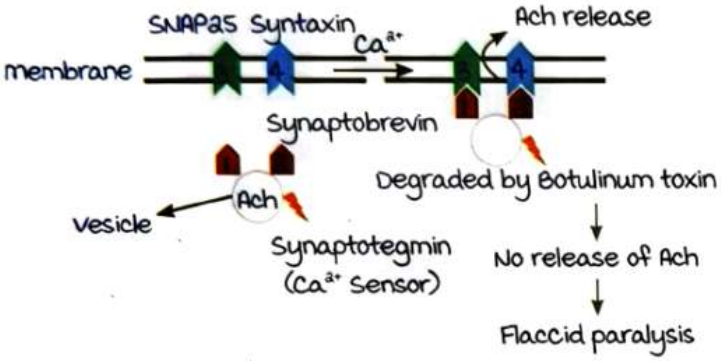
----- Active space -----

CLASS I :

Acetylcholine :

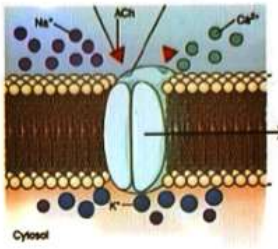
ACh release :

Acetylcholine release : Exocytosis by SNARE proteins & Ca^{2+}



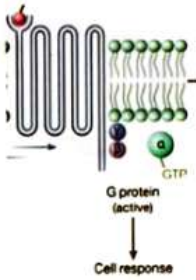
Receptors :

1. Nicotinic Ach receptor.



Ion channel : Ionotrope (Fast transmission)

2. Muscarinic Ach receptor.



G protein-coupled receptor : metabotropic (Slow transmission)

Location and function of acetylcholine :

1. Neuromuscular junction : Muscle contraction.
2. Nucleus basalis of meynert : Reduced level in Alzeihmer.
3. REM ON neurons : Sleep.
4. Sweat glands : Sympathetic cholinergic neurons.

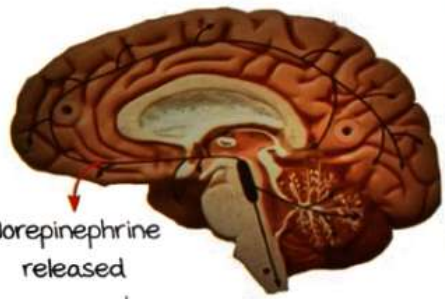
CLASS II : BIOGENIC AMINES.

Norepinephrine :

Synthesized from dopamine.

Location : Locus coeruleus.

Function : Promote wakefulness.



Norepinephrine released

↓

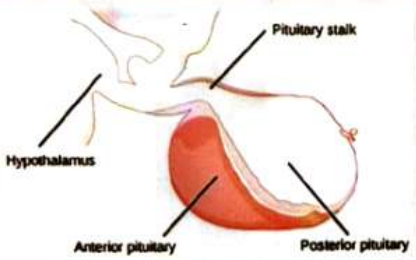
Widespread brain area activation

↓

Awake/arousal state attained

Dopamine :

Pathways :

	Nigrostriatal	mesolimbic	mesocortical	Tuberoinfundibular
Location	Basal ganglia (Substantia nigra)	Ventral tegmental area, Nucleus accumbens	Prefrontal cortex	
Function	Motor movement	• Reward pathway. • Addiction.	• Motivation. • Emotional regulation. • Decision making. • Memory.	Inhibit release of prolactin
Applied aspect	↓ level of dopamine : Bradykinesia in Parkinson's disease	Abnormal dopamine pathway at Nucleus Accumbens : Addiction	–	Dopamine : Physiologic inhibition of prolactin

Feedback

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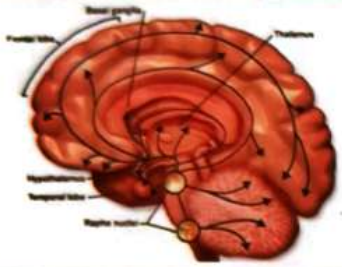
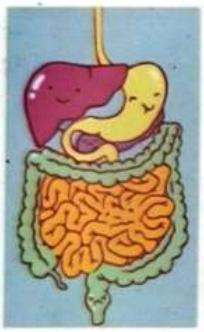
Receptors :

----- Active space -----

D1, D2, D3, D4, D5 : G-Protein-coupled receptors (GPCR).

Serotonin :

Released from :

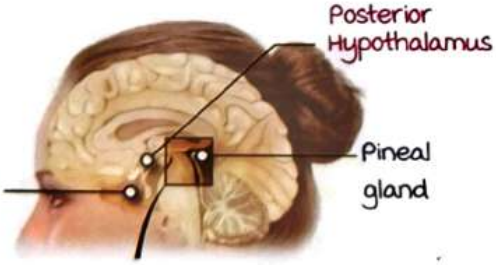
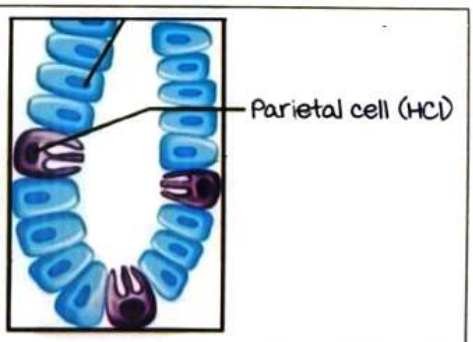
Location	Function
	Awake state
	Regulate peristalsis
	Aggregation

Receptors : 5-hydroxytryptamine receptors : SHT1 to SHT7.

	Type	Location	Function
SHT 2a	GPCR	Platelet	Platelet aggregation
SHT 2c		GIT	↓ food intake (Regulate satiety)
SHT 4		GIT	Peristalsis
SHT 6,7		Limbic system	
SHT 3	Ionotropic receptor	Area postrema	Vomiting

Histamine :

Released from :

Location		
Function	Promote wakefulness	↑ Acid production

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----- Active space -----

CLASS III : AMINO ACID.

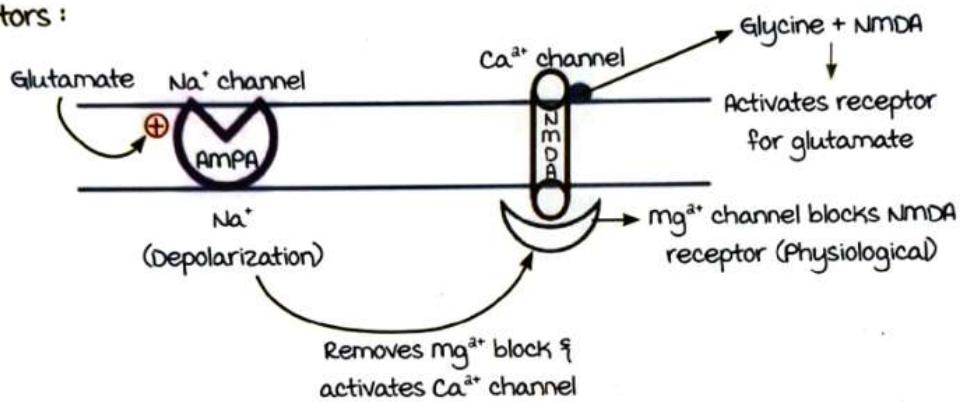
Glutamate :

- major excitatory neurotransmitters.

Released from :

Location	Function
Hippocampus, subthalamic nucleus	memory
A-delta fibres	Fast pain

Receptors :



Clinical aspect :

Excitotoxicity

Seen in amyotrophic lateral sclerosis (ALS).

Mechanism of action (MOA) : ↑ Ca2+ influx

↓

Cell death

Rx : Riluzole (NMDA blocker)

Note :

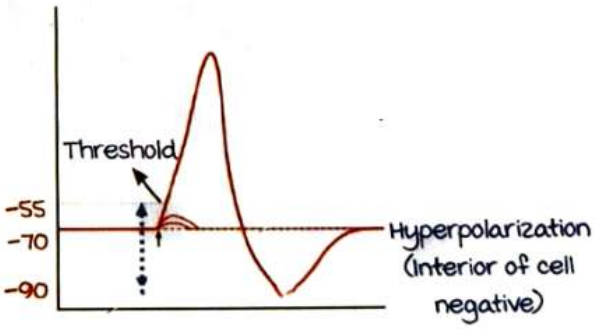
Subthalamic nucleus : Only excitatory nucleus of basal ganglia.

GABA :

- major inhibitory neurotransmitter.
- Location : Basal ganglia → caudate & putamen (Striatum).
- Function : Inhibit involuntary movements.
- MOA : Acts in hyperpolarization phase of action potential.
- Applied aspect : Loss of GABA in striatum

  ↓

  Huntington's chorea (Involuntary movement).

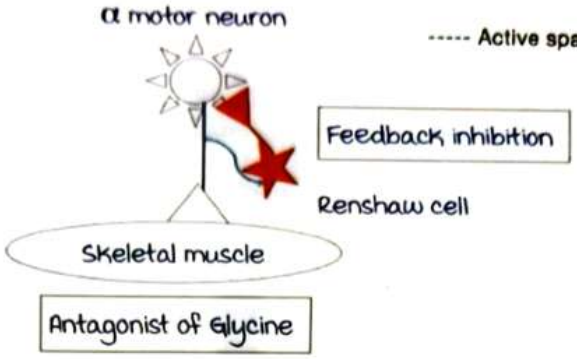


Note : GABA is inhibitory by enhancing hyperpolarization.

Feedback

Glycine :

- Function : Both excitatory & inhibitory.
- Location : α motor neuron.
- MOA : Feedback inhibition/renshaw cell inhibition.



Clinical aspect :

Strychnine → Inhibits the release of glycine

Tetanospasmin → Inhibits the release of GABA

→ Causes spastic paralysis

(D/t uninhibited stimulation of α motor neurons → Sustained contractions).

CLASS IV : GASES.

Nitric oxide :

Source : Arginine.

Location : Hippocampus.

MOA :

- Requires cGMP for all action.
- ↑ glutamate release from presynaptic neuron.

Function : Learning and memory.

Carbon monoxide :

Source :

Heme $\xrightarrow{\text{Heme oxygenase 2}}$ CO

Function :

- Olfaction.
- Pain transmission.
- Learning and memory.

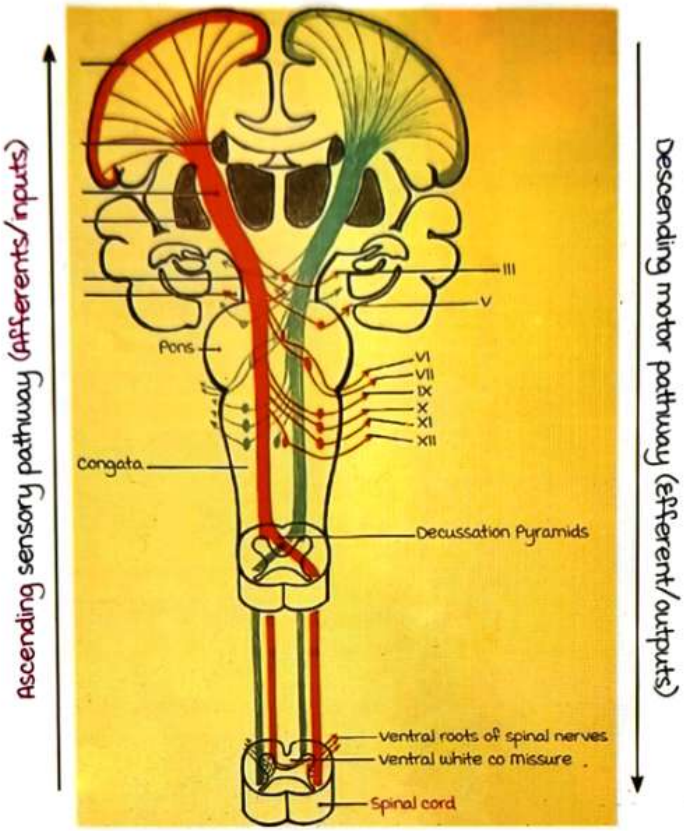
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SENSORY RECEPTORS

NEUROPHYSIOLOGY

- Sensory physiology.
- motor physiology.
- Higher mental functions.



Sensory Physiology

00:01:04

SENSES

General/somatic senses :

- Touch.
- Pain.
- Temperature.

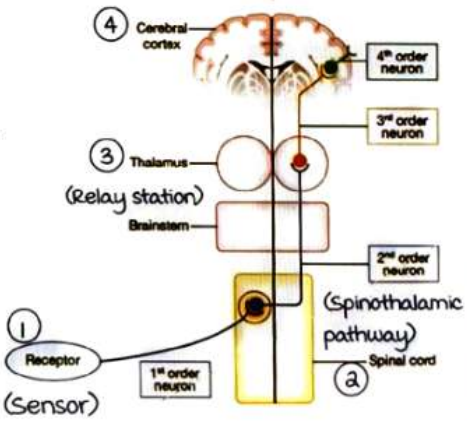
} Receptor : Skin.

Special senses :

- Vision.
- Hearing.
- Smell.
- Taste.
- Balance.

} Receptor : CNS.

SENSORY PATHWAY



Sensory pathways- Hierarchy Bottom_UP

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SENSORY RECEPTORS

----- Active space -----

Classification :

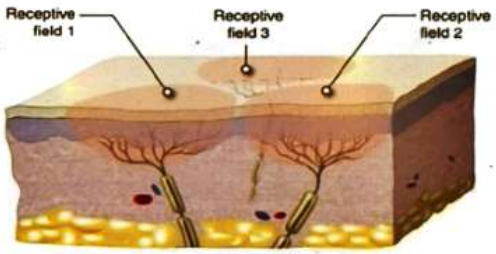
Receptor	Example
mechanoreceptors	Touch receptor
Nociceptors	Pain receptor
Thermoreceptors	Receptors sensing hot & cold
Proprioceptors	Joint position sensing receptor
Exteroceptors	Touch receptors (Skin responding to external stimuli)
Interoceptors	Osmoreceptors (Responds to osmotic changes in body)
Telereceptors	Photoreceptors (Sense light from distant stimuli)

Properties : (mnemonic : MILD)

1. Modality : Specific sense activating a receptor.
2. Intensity : Intensity ∝ frequency of action potential (Amplitude remains constant : All or none law).
3. Location :

   Receptive field : Area where the stimulus activates the receptor.

   Receptor density : Number of receptors in a receptive field.



   - Two-point discrimination : Minimum distance between 2 points at which they can be seen separately.
     - Receptor density $\propto \frac{1}{\text{Two point discrimination}}$
     - Two point discrimination : Maximum at lips, fingertips. Minimum at back region.
4. Duration :

   Time gap between start & end of receptor response.

General Senses 00:20:00

TOUCH

Touch receptors :

Location : Skin.

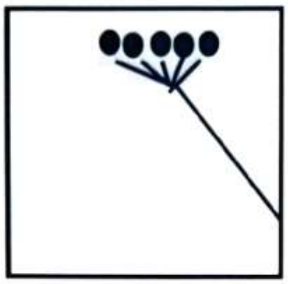
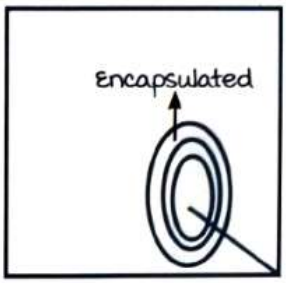
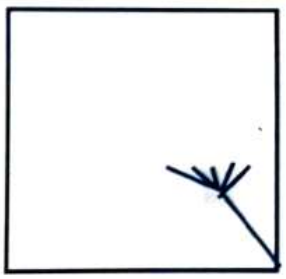
1st order neuron : Aβ afferent (Large, myelinated).

Feedback

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----- Active space ----- Types :

				
	Meissner's corpuscles	Merkel cells	Pacinian corpuscles (Largest)	Ruffini's endings
Location	Superficial (Found in lips, fingertips)	Superficial	Deep	Deep (Joint capsule receptor)
Function	• Tap. • Vibration (Low frequency 5 – 40 Hz).	• Touch. • Sensitive to edges & corners (Braille reading). • Sustained pressure.	• Pressure sensor. • Vibration (High frequency 50-500Hz).	• Skin stretch. • Vibrations.

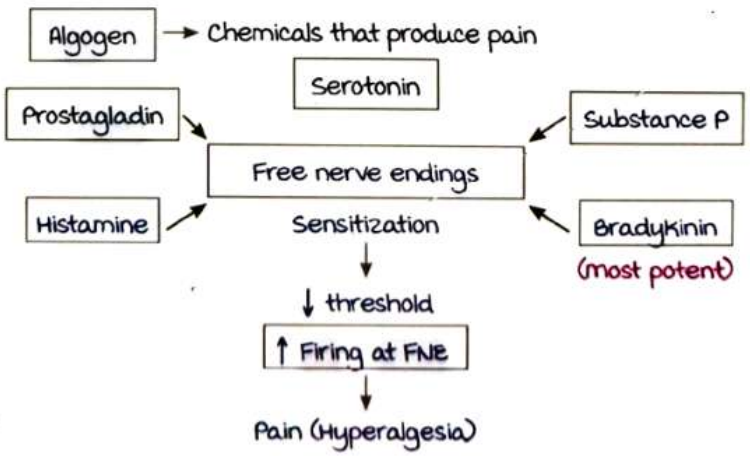
PAIN

Receptors : Free nerve endings of Aδ & C fibers.

	Aδ fibers (myelinated)	C fibers (unmyelinated)
Causes	1st pain/fast pain/epicritic pain	2nd pain/slow pain/protopathic pain
Pathway	Neospinothalamic	Paleospinothalamic
Neurotransmitter	Glutamate	Substance P

Somatic pain :

1. Hyperalgesia :
   - Seen in inflammatory conditions.



Note :

- Allodynia : Non nociceptive stimulus → Perceived as pain.

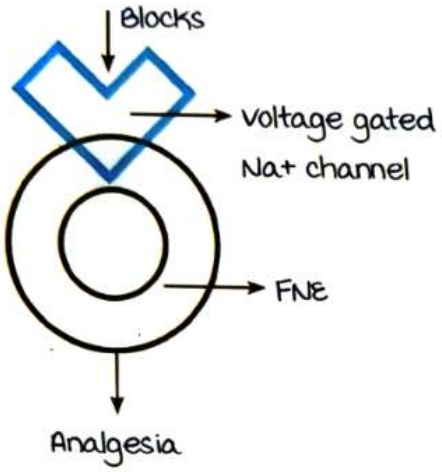
Feedback

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Clinical Aspect :

----- Active space -----

1. Local anesthetics (Eg : Lignocaine) :



2. Congenital insensitivity to pain syndrome (CIPS) :

Congenital absence of voltage gated Na+ channel.

2. Analgesia :

Type	Endogenous opioid analgesic system	Stress induced analgesic system
Example	Acupuncture therapy	Fire walking
MOA	Activation of periaqueductal gray matter (PAG) ↓ Release endogenous morphines (Eg : Endorphins, enkephalins, dynorphins)	Activation of PAG (D/t stress) ↓ Release of endogenous cannabinoids (Eg : Anandamide)
	Pain relief	

Clinical Aspect :

Massaging & transcutaneous electrical nerve stimulation (TENS).

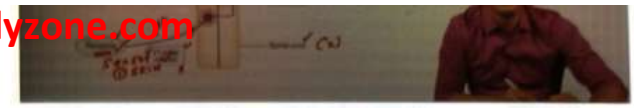




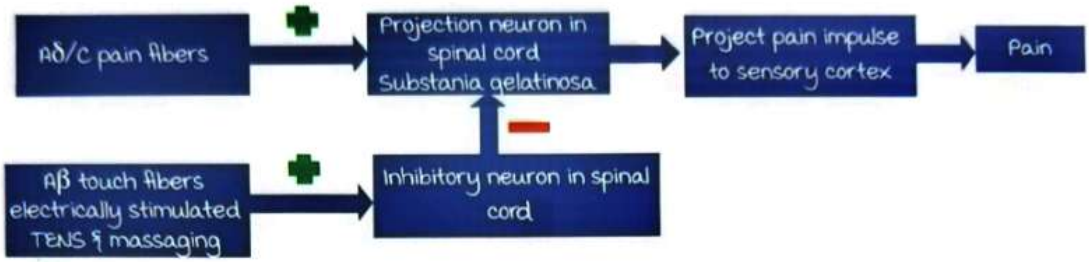
MOA : By gate control theory (Non painful sensations can override & reduce painful sensations).

Feedback

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----- Active space -----



Visceral Pain :

- Unpleasant, poorly localized, referred pain.
- Associated with nausea, vomiting & sweating.
- Conveyed via C fibers.
- Structures : Pain sensitive :
  - Liver capsule.
  - Parietal pleura.

  Pain insensitive :
  - Alveoli.

TEMPERATURE

Receptor :

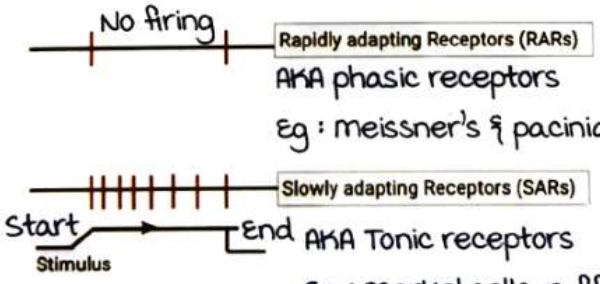


Thermoreceptor

	Cold sensing receptors number >> Warm sensing receptor	
Fibers	Aδ & C fibers	C fibers
Receptor	Transient receptor potential menthol (TRPm)	Transient receptor potential vanilloid (TRPV)
Temperature	10-24°C	30-45°C
Nociception	<5°C	>45°C

Note : Normal body temperature detected by TRPV3 receptor.

Adaptation of receptors :



AKA phasic receptors

Eg : meissner's & pacinian corpuscles

AKA Tonic receptors

Eg : merkel cells, ruffini's ending, free nerve ending.

Feedback